

Security Alert Monitoring & Incident Response Report

Future Interns – Cyber Security Internship (Task 2)

Name of Candidate: Aditya Choudhari
Internship Provider: Future Interns
Domain: Cyber Security (SOC Operations)
Date of Execution: 26 December 2025
Tool Used: Splunk Enterprise
Log Source: SOC_Task2_Sample_Logs.txt
Environment: Windows (Localhost)

1. Introduction

Security Operations Centers (SOC) play a critical role in monitoring, detecting, and responding to cybersecurity threats in real time. SOC analysts continuously analyze logs and alerts generated by systems and networks to identify suspicious activities and prevent potential security incidents.

This task simulates real-world SOC operations by analyzing system and network logs using a SIEM tool. The objective of this task was to monitor security alerts, identify potential threats, classify incidents based on severity, and propose appropriate incident response actions.

2. Objective

The main objectives of this task were:

- To gain hands-on experience with a SIEM tool
 - To analyze security logs and identify suspicious activities
 - To classify incidents based on severity (High, Medium, Low)
 - To document incident response actions and recommendations
 - To understand the workflow of a SOC analyst
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3. Tool Used

Splunk Enterprise

Splunk Enterprise is a powerful SIEM platform used for collecting, indexing, and analyzing machine-generated data. In this task, Splunk was used to ingest simulated SOC logs and perform keyword-based searches to detect security incidents.

4. Log Source Description

The log file [SOC Task2 Sample Logs.txt](#) contained simulated security events, including:

- Authentication logs (successful and failed logins)
- Network connection attempts
- File access activities
- Malware detection alerts

The logs were ingested as **unstructured text (misc_text)**, and keyword-based searches were used for incident detection.

5. Incident Detection Methodology

The following approach was used:

1. Upload logs into Splunk
 2. Perform keyword-based searches to identify suspicious activity
 3. Analyze patterns such as repeated failures, malware alerts, and unusual access
 4. Classify incidents by severity
 5. Document impact and response actions
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6. Stakeholder Notification Draft

Subject: Security Incident Alert – Malware Detection Identified

Dear Management Team,

During routine security monitoring activities, a security incident related to malware detection was identified through the SIEM monitoring system. The incident has been classified as High Severity due to the presence of malicious indicators such as Trojan and Rootkit signatures within the system logs.

Immediate response actions were initiated, including identification of affected systems, enhanced monitoring, and recommendations for malware removal and system isolation to prevent further impact.

Further analysis is ongoing to ensure that no additional systems are affected and to prevent recurrence of similar incidents. A detailed incident response report has been prepared and is available for review.

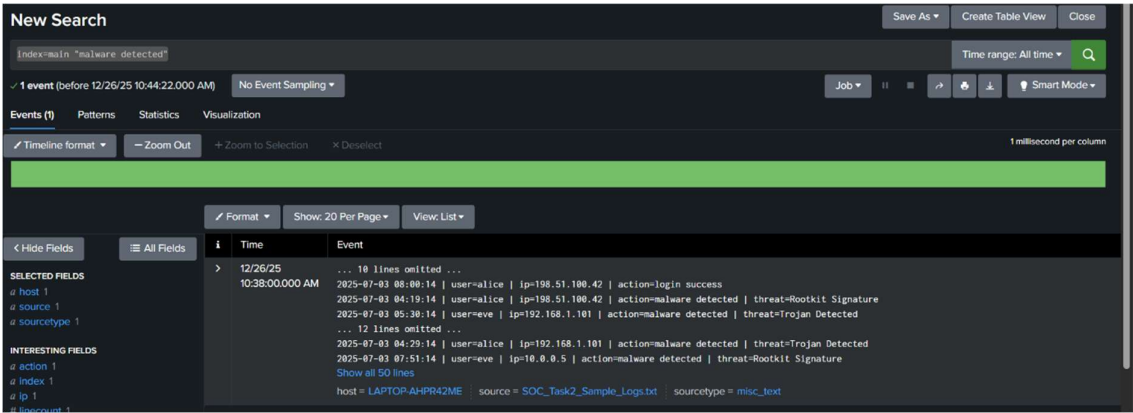
Please let us know if any additional actions or clarifications are required.

Regards,
Aditya Choudhari
SOC Analyst Intern
Future Interns

6. Identified Security Incidents

Incident 1: Malware Infection Detected

- **Search Query Used:**
`index=main "malware detected"`
- **Description:**
Multiple malware alerts were detected in the logs, including threats such as *Trojan Detected* and *Rootkit Signature*. These alerts indicate that malicious software was present on the system.
- **Severity: High**
- **Impact:**
Malware infections can lead to data theft, system compromise, and unauthorized access.
- **Recommended Response:**
 - Isolate the affected systems
 - Run full antivirus and malware scans
 - Remove infected files
 - Apply security patches and updates



Incident 2: Multiple Failed Login Attempts

- **Search Query Used:**
`index=main "login failed"`
- **Description:**
Failed login attempts were detected, indicating possible brute-force or credential-stuffing attacks.
- **Severity: Medium**
- **Impact:**
Repeated failed logins may result in account compromise if successful.
- **Recommended Response:**
 - Enable account lockout policies
 - Enforce strong password requirements
 - Monitor login attempts and block suspicious Ips

New Search

Save

index=main "login failed"

✓ 1 event (before 12/26/25 10:46:15.000 AM)

No Event Sampling

Job

Events (1)

Patterns

Statistics

Visualization

Timeline format

Zoom Out

+ Zoom to Selection

× Deselect

Format

Show: 20 Per Page

View: List

< Hide Fields

All Fields

SELECTED FIELDS

host 1

source 1

sourcetype 1

INTERESTING FIELDS

action 1

index 1

i

Time

Event

>

12/26/25

10:38:00.000 AM

... 15 lines omitted ...

2025-07-03 07:02:14 | user=alice | ip=203.0.113.77 | action=login failed

2025-07-03 04:18:14 | user=bob | ip=198.51.100.42 | action=login success

2025-07-03 09:02:14 | user=david | ip=203.0.113.77 | action=login failed

2025-07-03 09:07:14 | user=eve | ip=203.0.113.77 | action=login success

2025-07-03 04:47:14 | user=bob | ip=10.0.0.5 | action=login failed

Show all 50 lines

host = LAPTOP-AHPR42ME | source = SOC_Task2_Sample_Logs.txt | sourcetype = misc_text

Incident 3: Suspicious IP Address Activity

- **Search Query Used:**
`index=main "203.0.113.77"`
- **Description:**
The same external IP address was involved in multiple activities, including login attempts and access events, indicating suspicious behaviour.
- **Severity: Medium**
- **Impact:**
Repeated access attempts from a single IP may indicate reconnaissance or attack preparation.
- **Recommended Response:**
 - Block or restrict the suspicious IP
 - Monitor network traffic
 - Implement firewall and IDS/IPS rules

New Search

Save

index=main "203.0.113.77"

✓ 1 event (before 12/26/25 10:48:15.000 AM)

No Event Sampling

Job

Events (1)

Patterns

Statistics

Visualization

Timeline format

Zoom Out

+ Zoom to Selection

X Deselect

Format

Show: 20 Per Page

View: List

< Hide Fields

All Fields

SELECTED FIELDS

host 1

source 1

sourcetype 1

INTERESTING FIELDS

action 1

index 1

ip 1

linecount 1

i

Time

Event

>

12/26/25
10:38:00.000 AM

... 13 lines omitted ...

2025-07-03 06:10:14 | user=david | ip=203.0.113.77 | action=file accessed

2025-07-03 05:42:14 | user=eve | ip=203.0.113.77 | action=malware detected | threat=Trojan Detected

2025-07-03 07:02:14 | user=alice | ip=203.0.113.77 | action=login failed

... 1 line omitted ...

2025-07-03 09:02:14 | user=david | ip=203.0.113.77 | action=login failed

... 23 lines omitted ...

2025-07-03 05:06:14 | user=bob | ip=203.0.113.77 | action=malware detected | threat=Worm Infection Attempt

Show all 50 lines

host = LAPTOP-AHPR42ME | source = SOC_Task2_Sample_Logs.txt | sourcetype = misc_text

Incident 4: Unusual File Access Activity

- **Search Query Used:**
`index=main "file accessed"`
- **Description:**
File access events were detected that may indicate unauthorized or unusual access to sensitive files.
- **Severity: Low**
- **Impact:**
Could lead to data leakage if sensitive files are accessed without authorization.
- **Recommended Response:**
 - Review user permissions
 - Implement access control policies
 - Enable file activity monitoring

New Search

index=main "file accessed"

✓ 1 event (before 12/26/25 10:49:16.000 AM) No Event Sampling

Events (1) Patterns Statistics Visualization

Timeline format Zoom Out + Zoom to Selection × Deselect

< Hide Fields

All Fields

SELECTED FIELDS

a host 1

a source 1

a sourcetype 1

INTERESTING FIELDS

a action 1

a index 1

a ip 1

linecount 1

a punct 1

a splunk_server 1

Format

Show: 20 Per Page

View: List

i	Time	Event
>	12/26/25 10:38:00.000 AM	... 32 lines omitted ... 2025-07-03 09:10:14 user=bob ip=198.51.100.42 action=file accessed ... 1 line omitted ... 2025-07-03 08:31:14 user=eve ip=203.0.113.77 action=file accessed ... 10 lines omitted ... 2025-07-03 08:42:14 user=eve ip=172.16.0.3 action=file accessed ... 1 line omitted ... 2025-07-03 04:53:14 user=alice ip=203.0.113.77 action=file accessed ... 1 line omitted ... 2025-07-03 05:44:14 user=bob ip=198.51.100.42 action=file accessed Show all 50 lines host = LAPTOP-AHPR42ME source = SOC_Task2_Sample_Logs.txt sourcetype = misc_text

7. Incident Severity Classification Summary

Incident	Description	Severity
Malware Infection	Malware detected in system logs	High
Failed Login Attempts	Possible brute-force attack	Medium
Suspicious IP Activity	Repeated actions from one IP	Medium
Unusual File Access	Possible unauthorized access	Low

8. Challenges Faced

- Understanding SIEM workflows and navigation within Splunk
 - Working with unstructured logs and performing keyword-based searches
 - Correlating events to accurately identify security incidents
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9. Learning Outcomes

Through this task, I learned:

- Basics of SOC operations and alert monitoring
 - How to use a SIEM tool for log analysis
 - Incident classification and severity assessment
 - Importance of timely incident response
 - Professional documentation of security incidents
-

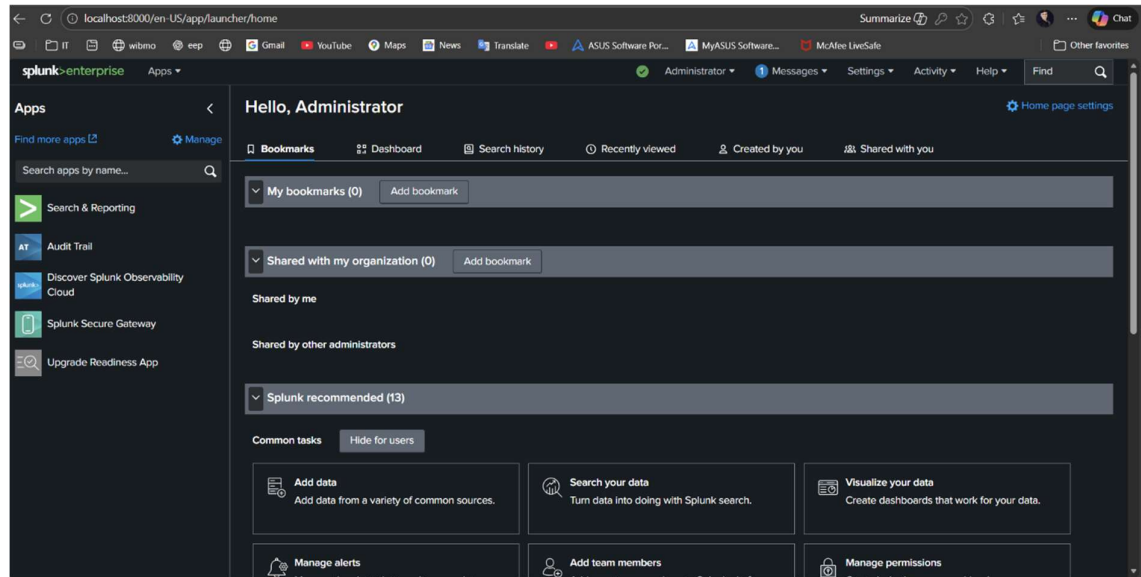
10. Conclusion

This task provided valuable hands-on experience in security alert monitoring and incident response. By analyzing logs using Splunk, multiple security incidents were successfully identified and classified. The exercise highlighted the importance of continuous monitoring, timely detection, and proper response to cybersecurity threats. Implementing the recommended security measures can significantly improve an organization's security posture.

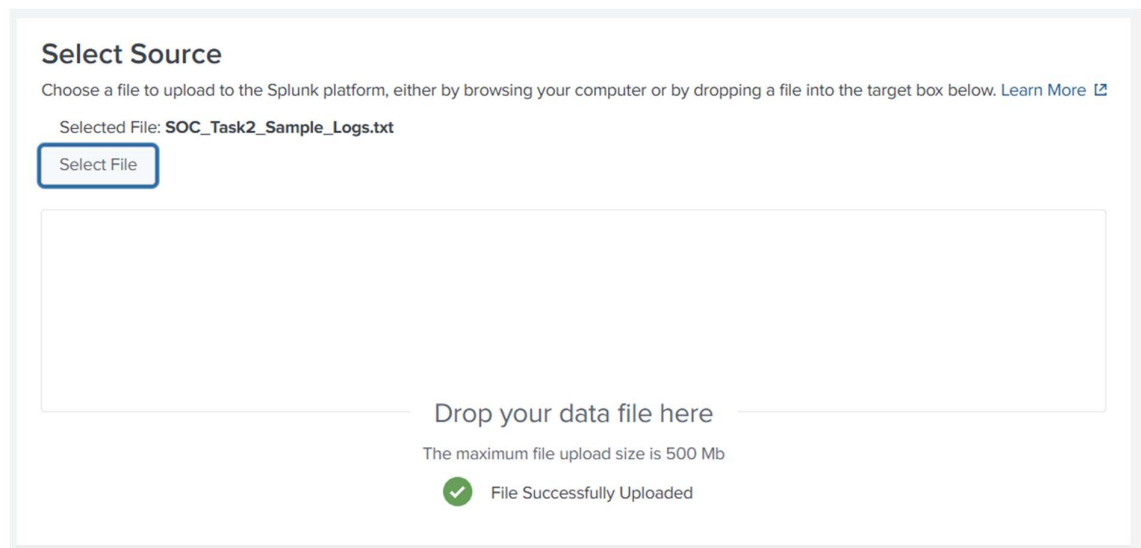
11. Screenshots & Evidence

Screenshots were captured during the log analysis process using Splunk Enterprise to support and validate the identified security incidents. The evidence includes screenshots of SIEM search queries and corresponding results highlighting suspicious activities such as malware detection, failed login attempts, suspicious IP behaviour, and unusual file access events.

1.



2.



Add Data

Select Source
→
Set Source Type
→
Input Settings
→
Review
→
Done

< Back
Next >

Set Source Type

This page lets you see how the Splunk platform sees your data before indexing. If the events look correct and have the right timestamps, click "Next" to proceed. If not, use the options below to define proper event breaks and timestamps. If you cannot find an appropriate source type for your data, create a new one by clicking "Save As".

Source: **SOC_Task2_Sample_Logs.txt**

Source type: Select Source Type ▾

Save As

[View Event Summary](#)

✓ Format ▾ Select... ▾ Select... ▾

< Prev 1 2 3 Next >

	Time	Event
1	7/3/25 6:13:14.000 AM	2025-07-03 06:13:14 user=charlie ip=10.0.0.5 action=connection attempt
2	7/3/25 8:20:14.000 AM	2025-07-03 08:20:14 user=charlie ip=192.168.1.101 action=connection attempt
3	7/3/25 5:04:14.000 AM	2025-07-03 05:04:14 user=bob ip=192.168.1.101 action=login success
4	7/3/25 6:01:14.000 AM	2025-07-03 06:01:14 user=bob ip=172.16.0.3 action=file accessed
5	7/3/25 5:18:14.000 AM	2025-07-03 05:18:14 user=charlie ip=172.16.0.3 action=login success
6	7/3/25 4:27:14.000 AM	2025-07-03 04:27:14 user=david ip=172.16.0.3 action=connection attempt
7	7/3/25 5:48:14.000 AM	2025-07-03 05:48:14 user=bob ip=10.0.0.5 action=malware detected threat=Trojan Detected

Add Data

Select Source

Set Source Type

Input Settings

Review

Done

< Back

Submit >

Review

Input Type	Uploaded File
File Name	SOC_Task2_Sample_Logs.txt
Source Type	misc_text
Host	LAPTOP-AHPR42ME
Index	main

slurp:enterprise
Apps
Administrator
Messages
Settings
Activity
Help
Find

Search
Analytics
Datasets
Reports
Alerts
Dashboards
Search & Reporting

New Search

Save As
Create Table View
Close

Time range: All time
Q

✓ 1 event (before 12/26/25 10:38:17:000 AM)
No Event Sampling
Job ▾
||
→
⬇
Smart Mode

Events (1)
Patterns
Statistics
Visualization

✓ Timeline format ▾
— Zoom Out
+ Zoom to Selection
× Deselect
1 millisecond per column

Format ▾
Show: 20 Per Page ▾
View: List ▾

< Hide Fields
≡ All Fields

SELECTED FIELDS

host 1
source 1
sourcetype 1

INTERESTING FIELDS

action 1
index 1
ip 1
linecount 1
punct 1
splunk_server 1
throat 1

#	Time	Event
>	12/26/25	
	10:38:00.000 AM	2025-07-03 06:13:14 user=charlie ip=10.0.0.5 action=connection attempt 2025-07-03 08:20:14 user=charlie ip=192.168.1.101 action=connection attempt 2025-07-03 05:04:14 user=bob ip=192.168.1.101 action=login success 2025-07-03 06:01:14 user=bob ip=172.16.0.3 action=file accessed 2025-07-03 05:18:14 user=charlie ip=172.16.0.3 action=login success Show all 50 lines
		host = LAPTOP-AHPR42ME source = SOC_Task2_Sample_Logs.txt sourcetype = misc.txt

6.

New Search Save As Create Table View Close

index=main "malware detected"

Time range: All time Q

✓ 1 event (before 12/26/25 10:44:22.000 AM) No Event Sampling Job || ≡ ↗ ⬇ ⬆ Smart Mode

Events (1) Patterns Statistics Visualization

Timeline format Zoom Out + Zoom to Selection X Deselect 1 millisecond per column

Format Show: 20 Per Page View: List

< Hide Fields All Fields

SELECTED FIELDS

- a host 1
- a source 1
- a sourcetype 1

INTERESTING FIELDS

- a action 1
- a index 1
- a ip 1
- a timestamp 1

#	Time	Event
>	12/26/25 10:38:00.000 AM	... 16 lines omitted ... 2025-07-03 06:00:14 user=alice ip=198.51.100.42 action=login success 2025-07-03 04:19:14 user=alice ip=198.51.100.42 action=malware detected threat=Rootkit Signature 2025-07-03 05:30:14 user=eve ip=192.168.1.101 action=malware detected threat=Trojan Detected ... 12 lines omitted ... 2025-07-03 04:29:14 user=alice ip=192.168.1.101 action=malware detected threat=Trojan Detected 2025-07-03 07:51:14 user=eve ip=10.0.0.5 action=malware detected threat=Rootkit Signature Show all 50 lines host = LAPTOP-AHPR42ME source = SOC_Task2_Sample_Logs.txt sourcetype = misc_text

7.

New Search Save

index=main "login failed"

✓ 1 event (before 12/26/25 10:46:15.000 AM) No Event Sampling Job || ≡

Events (1) Patterns Statistics Visualization

Timeline format Zoom Out + Zoom to Selection X Deselect

Format Show: 20 Per Page View: List

< Hide Fields All Fields

SELECTED FIELDS

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- a sourcetype 1

INTERESTING FIELDS

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- a index 1

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8.

New Search Save

index=main "203.0.113.77"

✓ 1 event (before 12/26/25 10:48:15.000 AM) No Event Sampling Job || ≡

Events (1) Patterns Statistics Visualization

Timeline format Zoom Out + Zoom to Selection X Deselect

Format Show: 20 Per Page View: List

< Hide Fields All Fields

SELECTED FIELDS

- a host 1
- a source 1
- a sourcetype 1

INTERESTING FIELDS

- a action 1
- a index 1
- a ip 1
- a timestamp 1

#	Time	Event
>	12/26/25 10:38:00.000 AM	... 13 lines omitted ... 2025-07-03 06:10:14 user=david ip=203.0.113.77 action=file accessed 2025-07-03 05:42:14 user=eve ip=203.0.113.77 action=malware detected threat=Trojan Detected 2025-07-03 07:02:14 user=alice ip=203.0.113.77 action=login failed ... 1 line omitted ... 2025-07-03 09:02:14 user=david ip=203.0.113.77 action=login failed ... 23 lines omitted ... 2025-07-03 05:06:14 user=bob ip=203.0.113.77 action=malware detected threat=Worm Infection Attempt Show all 50 lines host = LAPTOP-AHPR42ME source = SOC_Task2_Sample_Logs.txt sourcetype = misc_text

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New Search

index=main "file accessed"

✓ 1 event (before 12/26/25 10:49:16.000 AM) No Event Sampling ▼

Events (1) Patterns Statistics Visualization

Timeline format ▼ Zoom Out + Zoom to Selection × Deselect

Format ▼ Show: 20 Per Page ▼ View: List ▼

< Hide Fields All Fields

SELECTED FIELDS

a host 1

a source 1

a sourcetype 1

INTERESTING FIELDS

a action 1

a index 1

a ip 1

linecount 1

a punct 1

a splunk_server 1

i	Time	Event
>	12/26/25	... 32 lines omitted ...
	10:38:00.000 AM	2025-07-03 09:10:14 user=bob ip=198.51.100.42 action=file accessed
		... 1 line omitted ...
		2025-07-03 08:31:14 user=eve ip=203.0.113.77 action=file accessed
		... 10 lines omitted ...
		2025-07-03 08:42:14 user=eve ip=172.16.0.3 action=file accessed
		... 1 line omitted ...
		2025-07-03 04:53:14 user=alice ip=203.0.113.77 action=file accessed
		... 1 line omitted ...
		2025-07-03 05:44:14 user=bob ip=198.51.100.42 action=file accessed
		Show all 50 lines
		host = LAPTOP-AHPR42ME source = SOC_Task2_Sample_Logs.txt sourcetype = misc_text

